HW2: Long-Tailed Object Detection Report (TAICA/CVPDL)

Drone-based Long-Tailed Object Detection

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Abstract

We present a data-centric pipeline for a four-class long-tailed UAV object detection task (car, hov, person, motorcycle). Our approach combines (i) rigorous label conversion and cleaning (pixel-format boxes to YOLO format, duplicate merging, boundary clamping), (ii) small-object oriented *offline* augmentations (ASRZ, QRM, RCPB, HNT), (iii) mild re-sampling for minority classes, and (iv) training **YOLO11s from scratch** under the assignment constraints (no pretraining, dataset-only). At inference, a robust CSV builder de-normalizes predictions and enforces the competition submission specification. We report the full list of hyperparameters used in data preparation, augmentation, training, and inference to ensure exact reproducibility.

1 Introduction

Long-tailed detection in UAV imagery is challenging due to a high proportion of small objects, class imbalance, and background clutter. We adopt a data-first strategy: synthesize representative samples offline, then train a lightweight detector from scratch, followed by a deterministic post-processing routine that guarantees submission correctness.

Task and metric The task involves four classes and is evaluated by mean Average Precision at multiple IoU thresholds, $\text{mAP}_{50:95}$.

2 Dataset and Evaluation

Dataset. The training split contains labeled images (pixel-format boxes per line as `<class, left, top, w` the test split is unlabeled. Class IDs map to names as: 0: car, 1: hov, 2: person, 3: motorcycle.
Metric. We report $\text{mAP}_{50:95}$, i.e., AP averaged over IoU thresholds $\{0.50, 0.55, \ldots, 0.95\}$.

3 Method

We structure the pipeline as **Preprocessing** → **Offline Augmentation** → **Split / Resampling** → **Training** → **Inference / Submission**. Figure 1 overviews the workflow.

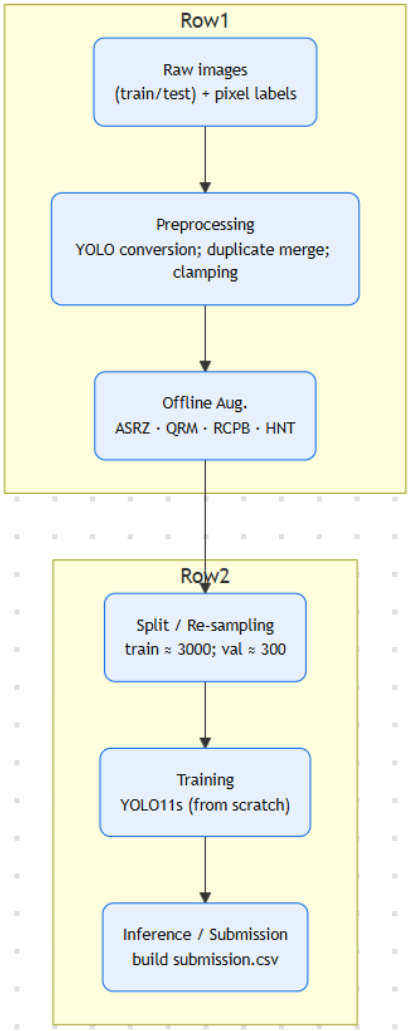

Figure 1: Pipeline overview. ASRZ: adaptive small-ROI zoom; QRM: quadrant ROI mosaic; RCPB: rectangular copy-paste bank; HNT: hard-negative tiles.

3.1 Preprocessing: Pixel to YOLO and Cleaning

Each raw label line is `<c, x, y, w, h>` in pixels. We perform: (1) **duplicate-box merge** for same-class pairs with IoU > 0.7 (keep the larger area), (2) **boundary clamping** to the image frame and tiny-box filtering (side ≤ 1 px discarded), (3) **spec conversion** to YOLO `<c xc yc wn hn>` (empty-label files allowed as hard negatives), and (4) a **stratified split** that keeps minority classes (hov/person) in train.

3.2 Offline Augmentation

We deploy four long-tail oriented augmentations. All outputs (images/labels) are written under `/aug/<type>/{images,labels}`.

ASRZ (Adaptive Small-ROI Zoom): select targets with YOLO area < 0.01, crop a square ROI with context, resize the short side to 960, and remap labels.

QRM (Quadrant ROI Mosaic): select one ROI from each of four images (favor small objects), place on a 1280×1280 canvas, and project intersections of boxes.

RCPB (Rect Copy-Paste Bank): build a class-wise patch bank from small/medium boxes and paste at random locations if IoU with existing boxes < 0.2; append labels.

HNT (Hard-Negative Tiles): crop random tiles with IoU < 0.02 to any GT and save them as empty-label negatives.

3.3 Complete Hyperparameters

Data construction and augmentation.

- **Random seed:** 42.
- **Train/val target sizes:** Train $T = 3000$, Val $V = 300$ (use all available if fewer).
- **Minority re-sampling:** images containing hov or person duplicated $\times 2$; others $\times 1$.
- **ASRZ:** `max_per_img=2`, `small_thr=0.01`, `zoom_min=2.2`, `zoom_max=3.0`, `out_short=960`.
- **QRM:** `out_size=1280`, choose 4 images per mosaic; total mosaics `qrm_N = min(300, len(train_list)`
- **RCPB bank:** `limit_per_cls=500`; collect boxes with YOLO area ≤ 0.03.
- **RCPB paste:** `trials=40`, `max_paste=4` per image, IoU threshold `iou_thr=0.2`, scale $\sim \mathcal{U}(0.8, 1.5)$.
- **HNT:** `tiles=2` per image; tile width $\sim U(0.25W, 0.5W)$, height $\sim U(0.25H, 0.5H)$; negative if max IoU < 0.02; label files left empty.
- **Label existence check:** only images with corresponding label files (possibly empty) are used for train/val lists.

4 Experiments and Results

Below we first enumerate the exact hyperparameters used for training and inference (Table 1), then provide an ablation template to log incremental gains for long-tail countermeasures (Table 2).

4.1 Training/Inference Settings

Table 1: Training and inference hyperparameters (exact values used)

Model	YOLO11s (from scratch; `yolo11s.yaml`, `pretrained=False`)
Train image size / batch / epochs	1152 / 8 / 100
Optimizer	SGD (`lr0=0.01`, `lrf=0.10`, `momentum=0.937`, `weight_decay=5e-4`)
Warmup / Reg.	`warmup_epochs=5.0`, `warmup_momentum=0.8`, `warmup_bias_lr=0.1`, `dropout=0.05`
Early stop / infra	`patience=20`, `cache=False`, `workers=8`, `seed=42`
Inference (test)	`imgsz=1280`, `conf=0.05`, `iou=0.6`, `max_det=3000`
Output saving	`save_txt=True`, `save_conf=True` (per-image label files)

4.2 Ablation (measured)

Table 2: Validation results (final configuration only). Overall $mAP_{50:95}$ and per-class $AP_{50:95}$.

Method	$mAP_{50:95}$	AP(car)	AP(hov)	AP(person)	AP(motorcycle)
Final (ours)	**0.287**	**0.413**	**0.391**	**0.134**	**0.211**

Validation: 300 images. Training: YOLO11s from scratch (`imgsz=1152`, batch 8, 100 epochs, SGD). Inference: `imgsz=1280`, `conf=0.05`, `iou=0.6`, `max_det=3000`. Overall: Precision 0.707, Recall 0.628, mAP_{50} 0.662, $mAP_{50:95}$ 0.287.

4.3 Qualitative Analysis

To complement the quantitative results, we provide a single qualitative montage summarizing typical scenes, small-object cases, and representative detections on the test set.

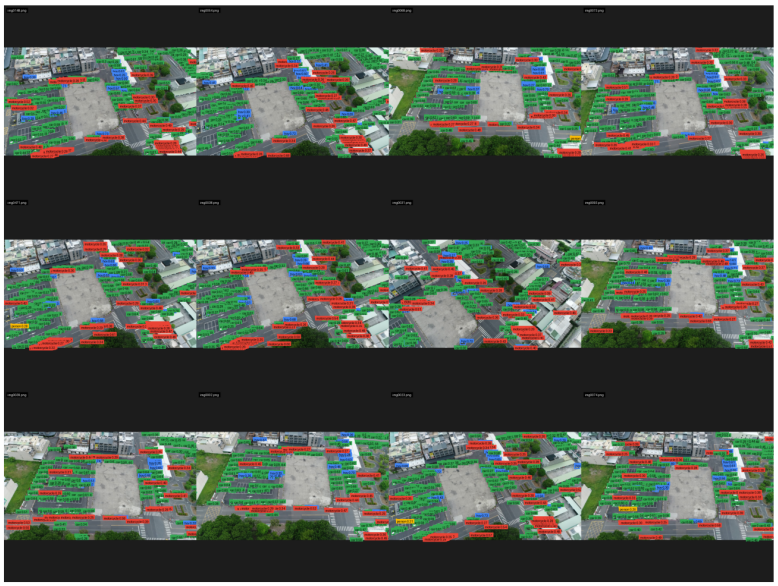

Figure 2: Qualitative overview: representative UAV scenes with detections (boxes and confidences). The montage includes small objects, occlusions, and cluttered backgrounds to illustrate typical success and failure patterns.

Observed patterns.

- Small objects (hov/person) benefit from offline augmentations (ASRZ/QRM), yielding clearer local context.
- Hard negatives (HNT) reduce background false positives (e.g., shadows, road markings).
- Remaining errors are dominated by tiny distant targets and severe occlusions; see Fig. 2.

5 Conclusion

We presented a data-centric pipeline for long-tailed UAV object detection that strictly adheres to the assignment constraints (no pretraining, dataset-only). On the 300-image validation split, our **final** configuration—offline augmentations (ASRZ, QRM, RCPB, HNT) plus mild minority re-sampling and tuned inference—achieves $\mathbf{mAP}_{50:95} = 0.287$ (overall precision **0.707**, recall **0.628**, $\text{mAP}_{50} = \mathbf{0.662}$). Per-class $\text{AP}_{50:95}$ indicates strongest performance on car (**0.413**) and hov (**0.391**), with motorcycle at **0.211** and person remaining the most challenging at **0.134**, consistent with the prevalence of tiny, partially occluded human instances in UAV scenes.

Qualitative inspection (Fig. 2) corroborates the quantitative trends: small-object zooming (ASRZ) and ROI mosaics (QRM) improve minority coverage, while hard-negative tiles suppress background false positives. Future work will target (i) stronger small-object handling (higher-resolution training with batch adjustment or tiling inference), (ii) class-aware thresholds and per-class NMS to stabilize minority recall, and (iii) dynamic loss re-weighting or sampling schedules to further close the person gap without sacrificing majority-class precision.

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